

Multiple Choice

1. **Answer:** A

Options: A) 3 gallons; B) 5 gallons; C) 8 gallons; D) 2 gallons; E) 1 gallon

Note: The correct answer is 3 gallons because, by setting up an equation based on the concentrations and total volumes of alcohol in the mixtures, it can be demonstrated that combining 3 gallons of the 74 percent alcohol liquid with 5 gallons of the 90 percent alcohol liquid yields a final mixture that is 84 percent alcohol.

2. **Answer:** A

Options: A) 14.15 M; B) 16.20 M; C) 15.00 M; D) 12.75 M; E) 9.95 M

Note: The correct answer of 14.15 M is derived from calculating the molarity by first determining the mass of HCl in a given volume of solution using its percentage concentration and density, then converting that mass to moles by dividing by the molar mass of HCl.

3. **Answer:** B

Options: A) 3.0×10^{-3} ; B) 2.0×10^{-3} ; C) 1.7×10^{-1} ; D) 1.5×10^{-2} ; E) 2.0×10^{-1}

Note: The ionization constant, K_a , for the weak monoprotic acid is calculated using the formula $K_a = \frac{[H^+][A^-]}{[HA]}$, where $[H^+]$ can be derived from the pH (which is 10⁻² for a pH of 2.0), leading to a K_a value of 2.0×10^{-3} .

3. **Answer:** A

Options: A) 1.2 dm³ mol⁻¹; B) 1.35 dm³ mol⁻¹; C) 2.0 dm³ mol⁻¹; D) 0.8 dm³ mol⁻¹; E) 1.8 dm³ mol⁻¹

Note: The correct answer is 1.2 dm³ mol⁻¹ because the molar volume calculated from the ideal gas law at 250 K and 15 atm is approximately 1.36 dm³ mol⁻¹, and since the actual molar volume is 12% smaller, it results in a value of about 1.2 dm³ mol⁻¹.

4. **Answer:** E

Options: A) 0.12 molal; B) 0.05 molal; C) 0.4 molal; D) 0.2 molal; E) 0.1 molal

Note: The correct answer is 0.1 molal because molality is calculated as the number of moles of solute (ethanol) divided by the mass of the solvent (water) in kilograms; with 2.3 g of ethanol corresponding to 0.05 moles and 500 g of water being 0.5 kg, the molality is 0.05 moles / 0.5 kg = 0.1 molal.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The root mean square speed of a gas is inversely proportional to the square root of its molar mass, and since the molar mass of hydrogen is significantly lower than

that of oxygen, the root mean square speed of oxygen can indeed be calculated to be around 400 m/sec at the same temperature as hydrogen.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the reaction of sulfuric acid with metallic calcium produces hydrogen gas in a stoichiometric ratio that does not yield 33.6 liters at STP, as the amount of gas produced depends on the specific quantities of reactants used, and 33.6 liters corresponds to the reaction of 1 mole of hydrogen gas, which cannot be achieved with the given reactants under these conditions.

8. **Answer:** False

Options: A) True; B) False

Note: The pKa of a weak monoprotic acid is not equal to its pH unless the acid is at its half-equivalence point in a titration, as the pH of 4.26 does not indicate the specific concentration of the acid and its dissociation equilibrium.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because an alcohol contains a hydroxyl (-OH) functional group, which includes an oxygen atom, thus disqualifying it from being classified purely as a hydrocarbon.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because it reflects the decreasing order of reducing ability based on the standard reduction potentials of the elements, where hydrogen has the highest reducing ability followed by nickel, cobalt, aluminum, sodium, and silver.

Long Form Response

11. **Answer:** To calculate the pH of a solution with a hydronium ion concentration of $10^{-11}M$, we use the formula $pH = -\log[H_3O^+]$. Substituting the given concentration, we find $pH = -\log(10^{-11}) = 11$. Since the pH scale ranges from 0 to 14, a pH of 11 indicates that the solution is basic.

Note: The answer correctly calculates the pH using the formula $pH = -\log[H_3O^+]$ for a hydronium ion concentration of $10^{-11}M$, resulting in a pH of 11, which indicates the solution is basic since $pH > 7$.

11. **Answer:** To calculate the cell potential for the reaction $Ni + Sn^{2+} \rightarrow Ni^{2+} + Sn$, we can use the Nernst equation, which incorporates the standard electrode potentials and the concentrations of the reactants and products. The standard reduction potential for $Sn^{2+} + 2e^- \rightarrow Sn$ is +0.14 volts, and for $Ni^{2+} + 2e^- \rightarrow Ni$, it is -0.23 volts. Therefore, the standard cell potential (E°) can be calculated as $E^\circ = E^\circ(\text{cathode}) - E^\circ(\text{anode}) = 0.14 - (-0.23) = 0.37 \text{ volts}$. However, given the specific concentrations of Ni^{2+} and Sn^{2+} , the Nernst equation yields a final cell potential of 0.15 volts. This positive value indicates that the reaction is spontaneous under the specified conditions, as a positive cell potential indicates a spontaneous reaction.

potential signifies a favorable free energy change.

Note: correct because it accurately applies the Nernst equation to calculate the standard cell potential by determining the standard reduction potentials for both half-reactions and correctly identifying the cathode and anode, which ultimately allows for assessing the spontaneity of the electrochemical reaction based on its positive cell potential.

End of Answer Key